

AI Career Platform - Complete Blueprint

Vision

Build an AI-powered Career Operating System aggregating jobs from multiple sources with AI-powered recommendations.

Core Features

Multi-source job aggregation; AI semantic search; Resume parsing; GitHub & portfolio analysis; Personalized recommendations; Skill-gap analysis; Salary insights; Company intelligence; One-click apply (supported platforms); Job alerts; Application tracking; AI career coach; Multi-language; Mobile-first; Global salary normalization.

Architecture

Frontend (Next.js) -> POST /api/v1/jobs/search -> Search Orchestrator -> Context Builder -> Job Aggregator (JobSpy, Adzuna, Jooble, Employer, Internal) -> Normalizer -> Duplicate Detection -> AI Enrichment -> Recommendation Engine -> Ranking -> Response. Storage: PostgreSQL, Redis, OpenSearch, pgvector.

Search Context

Query + Filters + User Profile + Resume + GitHub + Portfolio + Preferences + Career Goals + Search History + Saved Jobs + Applied Jobs + Market Insights.

AI Pipeline

Resume parsing, GitHub analysis, portfolio analysis, embeddings, semantic search, explainable ranking, interview probability, career roadmap.

Backend Modules

Auth, Users, Jobs, Search, Providers, Resume, GitHub, Portfolio, AI, Companies, Salary, Notifications, Analytics, Skills, Employers, Applications, Career, Admin.

Tech Stack

Next.js, React, Tailwind, NestJS, Prisma, PostgreSQL, Redis, OpenSearch, pgvector, BullMQ, Docker, Kubernetes.

Repositories

JobSpy: <https://github.com/speedyapply/JobSpy> ; Find Jobs: <https://github.com/AnumQ/find-jobs> ; Career Ops: <https://github.com/santifer/career-ops> ; AI Job Search: <https://github.com/maddykws/AI-Powered-Job-Search-Tool> ; OpenResume: <https://github.com/xitanggg/open-resume> ; Resume Matcher: <https://github.com/srbhr/Resume-Matcher> ; OpenSearch, Elasticsearch, Meilisearch, LangChain, LlamaIndex, Qdrant, pgvector, BullMQ, Scalar, Swagger UI.

Roadmap

Phase1 MVP; Phase2 AI personalization; Phase3 GitHub/Portfolio intelligence; Phase4 AI Career Coach & Employer AI.